8 CONTINUOUS IMPROVEMENT (80)	Total Marks 80.00

8.1 Actions taken based on the results of evaluation of each of the COs, POs & PSOs (40)	Total Marks 40.00

8.1.1 Actions Taken Based on the Results of Evaluation of the COs Attainment (20)	Institute Marks : 20.00

Documentary Evidence for Identification of Gaps in CO Attainment (20)

To ensure continuous improvement in Course Outcomes (COs), the Department of Computer Science and Engineering adopts a systematic micro-level gap analysis process. By critically examining CO attainment data, the department identifies weak areas in student performance and derives meaningful insights. Based on these observations, targeted corrective measures and strategic interventions are implemented to enhance teaching quality and overall learning effectiveness.

A. Evidence of Identification of Gaps in CO Attainment

The attainment of Course Outcomes (COs) serves as a key indicator of student learning and program effectiveness. Each course is designed with clearly defined COs that are aligned with Program Outcomes (POs) and institutional objectives. To maintain academic quality and meet accreditation requirements, CO attainment levels are periodically evaluated against predefined targets.

This document presents evidence of identified gaps in CO attainment across various courses, particularly where actual attainment levels fall below the set benchmarks. The analysis is supported by:

- CO attainment reports
- Internal and external assessment data
- Student feedback records
- Graphical analysis and performance trends

Table No. 8.1.1 provides detailed documentary evidence of gaps identified in CO attainment.

By consolidating assessment reports, feedback analysis, and action taken reports, the institution ensures a structured and transparent mechanism for identifying learning gaps. This process offers a comprehensive overview of student performance, highlighting both strengths and areas requiring improvement in:

- Conceptual understanding
- Problem-solving skills
- Practical application
- Communication abilities

Based on the identified gaps, the department implements structured remedial strategies such as:

- Interactive and student-centric teaching methodologies
- Enhanced laboratory practices
- Project-based learning approaches
- Workshops and skill development sessions
- Focused mentoring and academic support

Continuous monitoring of internal assessments, end-semester examinations, laboratory work, and project evaluations enables faculty to track student progress and assess the effectiveness of these interventions. This systematic approach ensures proper alignment of COs with POs and institutional goals, thereby fostering continuous improvement in the teaching–learning process and ensuring compliance with accreditation standards.

Fixing of Attainment Levels

The attainment levels are defined based on student performance as follows:

- If a student scores **70% or above**, the attainment level is **3**.
- If a student scores **between 60% and 69%**, the attainment level is **2**.
- If a student scores **between 50% and 59%**, the attainment level is **1**.
- If a student scores **below 50%**, the attainment level is **0**.

Table 8.1.1 CO Attainments for the Batch 2024-25

Note: The target value for each CO is 2.1.And the not attained co's showed in red color font in the following attainment table.

S. No	Course Code	Course Name	Year &Sem	CO Attainment					Attained
				CO1	CO2	CO3	CO4	CO5	
1	MA101BS	Matrices and Calculus	I Year I Sem	2.36	2.36	2.94	3.00	2.10	**Attained**
2	CH102BS	Engineering Chemistry	I Year I Sem	2.81	2.71	2.85	3.00	2.90	**Attained**
3	CS103ES	Programming for Problem Solving	I Year I Sem	2.40	2.32	2.37	2.26	2.29	**Attained**
4	EE104ES	Basic Electrical Engineering	I Year I Sem	2.27	2.55	2.10	2.10	3.00	**Attained**
5	EG105ES	Computer Aided Engineering Graphics	I Year I Sem	2.66	2.66	2.87	3.00	3.00	**Attained**
6	CS106ES	Elements of Computer Science & Engineering	I Year I Sem	2.72	2.72	2.86	3.00	3.00	**Attained**
7	CH107ES	Engineering Chemistry Laboratory	I Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
8	CS109ES	Programming for Problem Solving Laboratory	I Year I Sem	2.82	2.82	2.82	2.82	2.82	**Attained**
9	EE108ES	Basic Electrical Engineering Laboratory	I Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
10	HS110MC	Constitution of India	I Year I Sem	2.86	2.86	2.86	2.86	2.86	**Attained**
11	MA201ES	Ordinary Differential Equations and Vector Calculus	I Year II Sem	3.00	2.46	**1.59**	2.46	**1.37**	Not Attained

12	AP202ES	Applied Physics	I Year II Sem	2.98	2.53	2.99	2.98	2.97	**Attained**
13	ME203ES	Engineering Workshop	I Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
14	EN204ES	English for Skill Enhancement	I Year II Sem	2.77	2.54	2.69	2.61	2.61	**Attained**
15	EC205ES	Electronic Devices and Circuits	I Year II Sem	2.45	**1.79**	2.39	2.40	3.00	Not Attained
16	AP206ES	Applied Physics Laboratory	I Year II Sem	2.82	2.82	2.82	2.82	2.82	**Attained**
17	CS207ES	Python Programming Laboratory	I Year II Sem	**1.58**	**1.58**	**1.54**	**1.48**	**1.48**	Not Attained
18	EN208HS	English Language and Communication Skills Laboratory	I Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
19	CS209ES	IT Workshop	I Year II Sem	2.78	2.78	2.72	2.68	2.68	**Attained**
20	ES210MC	Environmental Science	I Year II Sem	2.77	2.54	2.69	2.61	2.61	**Attained**
21	PS301BS	Probability and Statistics	II Year I Sem	3.00	2.46	3.00	2.55	2.10	**Attained**
22	DE302ES	Digital Electronics	II Year I Sem	2.40	2.43	2.61	2.57	2.59	**Attained**
23	CS303PC	Data Structures	II Year I Sem	2.61	2.33	3.00	3.00	3.00	**Attained**
24	CS304PC	Computer Organization and Architecture	II Year	2.43	2.14	2.47	2.23	2.23	**Attained**

			I Sem						
25	CS305PC	Object Oriented Programming through Java	II Year I Sem	2.18	2.10	2.49	2.46	2.18	**Attained**
26	CS306PC	Data Structures Laboratory	II Year I Sem	2.81	2.81	2.86	2.91	2.91	**Attained**
27	CS307PC	Object Oriented Programming through Java Laboratory	II Year I Sem	2.68	2.68	2.72	2.78	2.78	**Attained**
28	HS309MC	Gender Sensitization	II Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
29	CS308PC	Skill Development Course(Data visualization-R Programming / Power BI)	II Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
30	MB401HS	Business Economics & Financial Analysis	II Year II Sem	2.55	2.55	2.00	**1.72**	2.55	Not Attained
31	CS402PC	Discrete Mathematics	II Year II Sem	2.91	2.46	2.98	2.39	2.94	**Attained**
32	CS403PC	Operating Systems	II Year II Sem	2.77	2.18	2.30	2.75	2.85	**Attained**
33	CS404PC	Database Management Systems	II Year II Sem	2.27	2.10	2.51	2.30	2.39	**Attained**
34	CS405PC	Software Engineering	II Year II Sem	2.10	2.10	2.47	2.49	2.34	**Attained**
35	CS406PC	Operating Systems Laboratory	II Year II Sem	2.90	2.90	2.89	2.90	2.90	**Attained**
36	CS407PC	Database Management Systems Laborator	II Year II Sem	2.91	2.91	2.91	2.91	2.91	**Attained**

37	CS408PW	Real-time Research Project/ Societal Related Project	II Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
38	HS411MC	Intellectual Property Rights	II Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
39	CS409PC	Skill Development Course(Node JS/ React JS/ Django)	II Year II Sem	2.91	2.91	2.91	2.91	2.91	**Attained**
40	CS501PC	Design and Analysis of Algorithms	III Year I Sem	3.00	2.75	2.85	2.92	2.87	**Attained**
41	CS502PC	Computer Networks	III Year I Sem	2.81	2.46	2.95	2.10	2.10	**Attained**
42	CS503PC	DevOps	III Year I Sem	3.00	2.92	3.00	3.00	3.00	**Attained**
43	CS513PE	Scripting Languages	III Year I Sem	2.31	2.38	2.85	2.46	2.38	**Attained**
44	CS515PE	Principles of programming languages02.26	III Year I Sem	2.26	2.43	2.52	2.61	2.61	**Attained**
45	CS523PE	Information Retrieval Systems	III Year I Sem	3.00	3.00	2.95	2.55	3.00	**Attained**
46	CS524PE	Distributed Databases	III Year I Sem	2.92	2.85	2.61	2.83	2.85	**Attained**
47	CS504PC	Computer Networks Laboratory	III Year I Sem	2.70	2.70	2.84	3.00	3.00	**Attained**
48	CS505PC	DevOps Laboratory	III Year I Sem	2.46	2.46	2.46	2.46	2.46	**Attained**

49	AE506HS	Advanced English Communication Skills Laboratory	III Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
50	CS507PC	Skill Development Course(UI designFlutter)	III Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
51	CS601PC	Machine Learning	III Year II Sem	3.00	3.00	3.00	2.55	2.55	**Attained**
52	CS602PC	Formal Languages and Automata Theory	III Year II Sem	2.36	2.36	2.41	2.55	2.42	**Attained**
53	CS603PC	Artificial Intelligence	III Year II Sem	2.69	2.15	2.61	2.15	2.61	**Attained**
54	CS632PE	Internet of Things	III Year II Sem	3.00	3.00	2.94	2.90	3.00	**Attained**
55	CS633PE	Data Analytics	III Year II Sem	3.00	3.00	2.61	2.61	2.61	**Attained**
56	CE611OE	Disaster preparedness and planning management	III Year II Sem	2.55	2.10	2.55	2.55	2.55	**Attained**
57	EE611OE	Renewable Energy Sources	III Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
58	CS604PC	Machine Learning Laboratory	III Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
59	CS605PC	Artificial Intelligence Laboratory	III Year II Sem	2.82	2.82	2.82	2.82	2.82	**Attained**
60	CS642PE	Internet of Things Laboratory	III Year II Sem	2.98	2.98	2.98	2.98	2.98	**Attained**
61	CS643PE	Data Analytics Laboratory	III Year	2.36	2.36	2.41	2.46	2.46	**Attained**

			II Sem						
62	CS606PW	Industrial Oriented Mini Project/ Internship/Skill Development Course (Big data-Spark)	III Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
63	MS701HS	Management Science	IV Year I Sem	2.94	2.52	2.12	2.52	2.52	**Attained**
64	CS702PC	Linux Programming	IV Year I Sem	2.30	2.69	2.45	2.58	2.79	**Attained**
65	CS731PE	Multimedia Computing	IV Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
66	CS732PE	Bigdata Analytics	IV Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
67	CS742PE	Cloud Computing	IV Year I Sem	2.71	2.41	2.74	3.00	3.00	**Attained**
68	CS743PE	Artificial Intelligence	IV Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
69	CE721OE	Disaster management	IV Year I Sem	2.58	3.00	3.00	3.00	3.00	**Attained**
70	CS703PC	Linux Programming Lab	IV Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
71	CS704PW	Mini project Evaluation	IV Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
72	CS705PW	Project Stage-I	IV Year I Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
73	CS852PE	Internet of Things	IV Year II Sem	2.45	2.54	2.41	2.48	2.45	**Attained**

74	CS831PE	Software project Management	IV Year II Sem	2.58	2.51	2.29	2.87	3.00	**Attained**
75	CS861PE	Mobile Computing	IV Year II Sem	2.53	2.45	2.53	2.53	2.53	**Attained**
76	CS862PE	Cyber Forensics	IV Year II Sem	2.16	2.16	2.16	2.16	2.16	**Attained**
77	CE831OE	Project Management	IV Year II Sem	3.00	3.00	2.58	3.00	2.58	**Attained**
78	CE832OE	Safety Engineering	IV Year II Sem	3.00	2.58	3.00	3.00	3.00	**Attained**
79	CS801PW	Seminar	IV Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
80	CS802PW	Project Stage-II	IV Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**

Table 8.1.2. Gap Analysis and Plan of implementation for the attainment of course outcomes (COs) for the Batch 2024-25

Course code MA201ES	**Course Title** Ordinary Differential Equations and Vector Calculus	**Attainment Levels of COs**					
		COs	1	2	3	4	5
		Attained	3.00	2.46	**1.59**	2.46	**1.37**

Gap Analysis:

CO3 (1.41) shows difficulty in higher-order differential equations, while CO5 (1.63) indicates challenges in applying vector calculus.

Plan for Implementation: Provide **assignment-based learning** with application-oriented questions

Implementation: 1. Conducted Continuous assessment through **quizzes, assignments.**

2.Introduced application-oriented examples and practice exercises.

Course code	Course Title	Attainment Levels of COs					
EC205ES	Electronic Devices and Circuits	**COs**	1	2	3	4	5
		Attained	2.45	**1.79**	2.39	2.40	3.00

Gap Analysis:

CO2: Significant gap (1.21), showing students face difficulty in understanding core device principles and circuit analysis

Plan for Implementation: Introduce **concept-based assignments** targeting weak.

Implementation: Scheduled **extra tutorial hours** for slow learners, especially for CO2.

Course code	Course Title	Attainment Levels of COs					
CS207ES	Python Programming Laboratory	**COs**	1	2	3	4	5
		Attained	**1.58**	**1.58**	**1.54**	**1.48**	**1.48**

Gap Analysis: All COs show significant gaps (>1.4), with highest in CO4 & CO5, indicating difficulties in advanced programming, weak fundamentals in CO1 & CO2, and poor problem-solving skills in CO3.

Plan for Implementation:

1. Conduct **bridge courses on basic programming concepts**

2. Introduce **weekly coding assignments and mini-program tasks**

Implementation:

1. Faculty monitored students individually during lab sessions.

Course code MB401HS	Course Title Business Economics & Financial Analysis	Attainment Levels of COs					
		COs	1	2	3	4	5
		Attained	2.55	2.55	2.00	**1.72**	2.55

Gap Analysis:

CO4: Highest gap (1.28), indicating lack of clarity in financial analysis and decision-making concepts

Plan for Implementation: Increase emphasis on **case studies and real-life business examples.**

Implementation: Encouraged **active participation in discussions and presentations.**

Table 8.1.2 CO Attainments for the Academic Year 2023-24

The target attainment level for each Course Outcome (CO) is set at **2.1**. The COs that have not aattained the target level are highlighted in **red color** in the attainment table for clear identification and further analysis..

S.No	Course Code	Course Name	Year &Sem	CO Attainment					Attainment
				CO1	CO2	CO3	CO4	CO5	
1	MA101BS	Matrices and Calculus	I Year I Sem	**2.40**	**2.40**	**2.39**	**2.40**	**2.40**	**Attained**
2	CH102BS	Engineering Chemistry	I Year I Sem	**2.37**	**2.37**	**2.35**	**2.37**	**2.37**	**Attained**
3	CS103ES	Programming for Problem Solving	I Year I Sem	**2.21**	**2.21**	**2.17**	**2.21**	**2.21**	**Attained**
4	EE104ES	Basic Electrical Engineering	I Year I Sem	**2.15**	**2.15**	**1.89**	**1.81**	**1.81**	**Not Attained**
5	EG105ES	Computer Aided Engineering Graphics	I Year I Sem	**2.15**	**2.15**	**2.28**	**2.46**	**2.46**	**Attained**

6	CS106ES	Elements of Computer Science & Engineering	I Year I Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
7	CH107ES	Engineering Chemistry Laboratory	I Year I Sem	**2.46**	**2.46**	**2.46**	**2.46**	**2.46**	**Attained**
8	CS109ES	Programming for Problem Solving Laboratory	I Year I Sem	**2.16**	**2.16**	**2.19**	**2.26**	**2.26**	**Attained**
9	EE108ES	Basic Electrical Engineering Laboratory	I Year I Sem	**2.46**	**2.46**	**2.46**	**2.46**	**2.46**	**Attained**
10	HS110MC	Constitution of India	I Year I Sem	**3.00**	**3.00**	**2.60**	**3.00**	**2.60**	**Attained**
11	MA201ES	Ordinary Differential Equations and Vector Calculus	I Year II Sem	**2.30**	**1.91**	**2.39**	**2.56**	**2.17**	**Not Attained**
12	AP202ES	Applied Physics	I Year II Sem	**2.39**	**2.39**	**2.74**	**2.10**	**2.48**	**Not Attained**
13	ME203ES	Engineering Workshop	I Year II Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
14	EN204ES	English for Skill Enhancement	I Year II Sem	**2.77**	**2.85**	**2.85**	**2.77**	**2.92**	**Attained**
15	EC205ES	Electronic Devices and Circuits	I Year II Sem	**1.91**	**2.30**	**1.96**	**2.87**	**2.87**	**Not Attained**
16	AP206ES	Applied Physics Laboratory	I Year II Sem	**2.80**	**2.80**	**2.78**	**2.80**	**2.80**	**Attained**
17	CS207ES	Python Programming Laboratory	I Year II Sem	**1.32**	**1.32**	**1.28**	**1.32**	**1.32**	**Not Attained**

18	EN208HS	English Language and Communication Skills Laboratory	I Year II Sem	1.38	1.38	1.38	1.38	1.38	Not Attained
19	CS209ES	IT Workshop	I Year II Sem	1.74	1.74	1.74	1.74	1.74	Not Attained
20	ES210MC	Environmental Science	I Year II Sem	1.38	1.38	1.38	1.38	1.38	Not Attained
21	PS301BS	Probability and Statistics	II Year I Sem	2.52	1.56	1.68	2.40	2.37	Not Attained
22	DE302ES	Digital Electronics	II Year I Sem	2.27	2.46	2.45	2.55	2.53	Attained
23	CS303PC	Data Structures	II Year I Sem	2.46	2.17	2.20	2.23	2.23	Attained
24	CS304PC	Computer Organization and Architecture	II Year I Sem	1.43	1.23	1.53	1.62	1.56	Not Attained
25	CS305PC	Object Oriented Programming through Java	II Year I Sem	2.27	2.27	2.42	2.41	2.32	Attained
26	CS306PC	Data Structures Laboratory	II Year I Sem	2.52	2.52	2.55	2.62	2.62	Attained
27	CS307PC	Object Oriented Programming through Java Laboratory	II Year I Sem	2.62	2.62	2.60	2.62	2.62	Attained
28	CS308PC	Skill Development Course(Data visualization-R Programming / Power BI)	II Year I Sem	2.72	2.72	2.71	2.72	2.72	Attained
29	MB401HS	Business Economics & Financial Analysis	II Year II Sem	2.94	1.63	2.94	2.01	2.92	Not Attained
30	CS402PC	Discrete Mathematics	II Year	2.9	2.54	2.07	2.61	2.31	Not Attained

			II Sem						
31	CS403PC	Operating Systems	II Year II Sem	**2.74**	**2.24**	**2.49**	**2.46**	**2.79**	**Attained**
32	CS404PC	Database Management Systems	II Year II Sem	**2.30**	**1.64**	**1.86**	**1.50**	**2.12**	**Not Attained**
33	CS405PC	Software Engineering	II Year II Sem	**2.35**	**2.35**	**2.44**	**2.72**	**2.29**	**Attained**
34	CS406PC	Operating Systems Laboratory	II Year II Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
35	CS407PC	Database Management Systems Laboratory	II Year II Sem	**2.80**	**2.80**	**2.73**	**2.70**	**2.70**	**Attained**
36	CS408PW	Real-time Research Project/ Societal Related Project	II Year II Sem	**2.46**	**2.46**	**2.16**	**2.46**	**2.46**	**Attained**
37	CS409PC	SkillDevelopment Course(Node JS/ React JS/ Django)	II Year II Sem	**2.62**	**2.62**	**2.60**	**2.62**	**2.62**	**Attained**
38	EC501PC	Computer Networks	III Year I Sem	**2.30**	**2.22**	**2.24**	**2.37**	**2.37**	**Attained**
39	CS502PC	Operating Systems	III Year I Sem	**2.88**	**2.88**	**2.89**	**2.90**	**2.87**	**Attained**
40	CS503PC	Compiler Design	III Year I Sem	**2.43**	**2.85**	**2.87**	**2.58**	**2.58**	**Attained**
41	CS504PC	Software Engineering	III Year I Sem	**2.90**	**3.00**	**2.93**	**3.00**	**3.00**	**Attained**
42	EC505PC	Microprocessors & Microcontrollers	III Year I Sem	**2.16**	**2.43**	**2.51**	**2.16**	**2.58**	**Attained**

43	CS104ES	Programming with Python	III Year I Sem	**2.51**	**2.70**	**2.80**	**2.58**	**2.58**	**Attained**
44	EN506HS	Advanced English Communication Skills Lab	III Year I Sem	**2.89**	**2.89**	**2.89**	**2.89**	**2.89**	**Attained**
45	CS507PC	Operating Systems and Computer Networks Lab	III Year I Sem	**1.94**	**1.94**	**2.05**	**2.16**	**2.16**	**Not Attained**
46	AE601HS	Managerial Economics and Financial Analysis	III Year II Sem	**3.00**	**2.45**	**3.00**	**2.58**	**2.58**	**Attained**
47	CS602PC	Data Warehousing and Data Mining	III Year II Sem	**2.78**	**3.00**	**2.87**	**3.00**	**2.58**	**Attained**
48	CS603PC	Object Oriented Analysis and Design	III Year II Sem	**2.58**	**2.01**	**3.00**	**2.58**	**2.58**	**Not Attained**
49	CS604PC	Web Technologies	III Year II Sem	**3.00**	**2.51**	**2.58**	**2.58**	**2.58**	**Attained**
50	CS622PE	Distributed Systems	III Year II Sem	**3.00**	**2.93**	**2.70**	**2.83**	**3.00**	**Attained**
51	CE612OE	Waste Management	III Year II Sem	**2.93**	**2.85**	**3.00**	**3.00**	**3.00**	**Attained**
52	CS605PC	Data Mining and Object Oriented Analysis and Design Lab	III Year II Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
53	CS606PC	Web Technologies Lab	III Year II Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
54	CS701PC	Management Science	IV Year I Sem	**2.12**	**2.12**	**1.70**	**1.64**	**2.03**	**Not Attained**
55	CS702PC	Linux Programming	IV Year	**2.31**	**2.96**	**2.96**	**2.48**	**2.45**	**Attained**

			I Sem						
56	CS731PE	Multimedia Computing	IV Year I Sem	**1.55**	**2.31**	**1.85**	**1.64**	**2.03**	**Not Attained**
57	CS732PE	BigData Analytics	IV Year I Sem	**2.04**	**2.54**	**2.54**	**2.06**	**2.03**	**Not Attained**
58	CS742PE	Cloud Computing	IV Year I Sem	**1.47**	**1.55**	**1.63**	**1.64**	**1.48**	**Not Attained**
59	CE721OE	Disaster management	IV Year I Sem	**2.46**	**2.96**	**2.89**	**2.48**	**2.45**	**Attained**
60	CS703PC	Linux Programming Lab	IV Year I Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
61	CS704PW	Mini project Evaluation	IV Year I Sem	**3.00**	**3.00**	**3.00**	**3.00**	**3.00**	**Attained**
62	CS705PW	Project Stage-I	IV Year I Sem	**3.00**	**3.00**	**2.60**			**Attained**
63	CS852PE	Internet of Things	IV Year II Sem	**2.58**	**3.00**	**2.58**	**2.58**	**2.58**	**Attained**
64	CS861PE	Mobile Computing	IV Year II Sem	**2.58**	**2.58**	**2.58**	**2.58**	**2.58**	**Attained**
65	CE831OE	Project Management	IV Year II Sem	**2.58**	**2.93**	**2.58**	**2.16**	**2.58**	**Attained**
66	CS801PW	Seminar	IV Year II Sem	3.00	3.00	3.00	3.00	3.00	**Attained**
67	CS802PW	Project Stage-II	IV Year II Sem			**3.00**	**3.00**	**3.00**	**Attained**

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
EE104ES	Basic Electrical Engineering	**Attained**	**2.15**	**2.15**	**1.89**	**1.81**	**1.81**

Gap Analysis: CO3, CO4, and CO5 have lower attainment compared to CO1 and CO2, indicating the need for better conceptual understanding and problem-solving skills in these areas..

Plan for Implementation: Focus on improving weak COs through extra practice sessions, concept-based teaching, and use of ICT tools. Provide additional assignments and tutorials.

Implementation: Conducted regular quizzes, and use interactive teaching methods to improve student performance in CO3, CO4, and CO5.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
MA201ES	Ordinary Differential Equations and Vector Calculus	**Attained**	**2.30**	**1.91**	**2.39**	**2.56**	**2.17**

Gap Analysis:

1.**CO2:** Conceptual difficulty / poor problem-solving skills

Plan for Implementation: Use **tutorial sheets with increasing difficulty levels**

Implementation: Conducted weekly quizzes

.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
AP202ES	Applied Physics	**Attained**	**2.39**	**2.39**	**2.74**	**2.10**	**2.48**

Gap Analysis:

1.**CO4:** Students show difficulty in understanding advanced concepts and their applications

Plan for Implementation: Use **application-oriented problems** and **real-life examples** to strengthen conceptual clarity.

Implementation: Conducted remedial classes and practice sessions with numerical problems..

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
EC205ES	Electronic Devices and Circuits	**Attained**	**1.91**	**2.30**	**1.96**	**2.87**	**2.87**

Gap Analysis:

1.CO1: Students have difficulty in understanding basic concepts and device fundamentals.

2.CO3: Students show weakness in problem-solving and circuit analysis.

Plan for Implementation: Use concept-oriented teaching with diagrams and simplified notes. Provide step-by-step problem-solving sessions and circuit analysis practice.

Implementation: Conducted remedial classes and explained concepts using device diagrams and solved previous question papers.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS207ES	Python Programming Laboratory	**Attained**	**1.32**	**1.32**	**1.28**	**1.32**	**1.32**

Gap Analysis:

Difficulty in basic programming concepts and applying concepts, Weak in problem-solving and writing programs.

Plan for Implementation: Use simple programs and demonstrations, Provide more lab practice, Assign more coding tasks and Provide debugging practice.

Implementation: Conducted extra lab sessions.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
EN208HS	English Language and Communication Skills Laboratory	**Attained**	**1.38**	**1.38**	**1.38**	**1.38**	**1.38**

Gap Analysis:

Students have low communication and language skills.

Plan for Implementation: Conduct speaking, listening, and practice activities

Implementation: Conducted communication practice sessions and lab activities.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS209ES	IT Workshop	**Attained**	**1.74**	**1.74**	**1.74**	**1.74**	**1.74**

Gap Analysis: Students have low practical skills in basic IT tools and operations..

Plan for Implementation: Provide hands-on practice and step-by-step guidance.

Implementation: Conducted lab exercises, demonstrations, and practice tasks.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
ES210MC	Environmental Science	**Attained**	**1.38**	**1.38**	**1.38**	**1.38**	**1.38**

Gap Analysis: Students have low understanding of environmental concepts and awareness

.

Plan for Implementation: Use real-life examples, case studies, and interactive learning.

Implementation: Conducted **discussions, assignments, and awareness activities**. Conducted **discussions, assignments, and awareness activities**.

Course code	Course Title	Attainment Levels of COs

Course code	Course Title	Attainment Levels of COs					
PS301BS	Probability and Science	**COs**	1	2	3	4	5
		Attained	**2.52**	**1.56**	**1.68**	**2.40**	**2.37**

Gap Analysis:

Students have difficulty in probability concepts and numerical problem-solving (CO2 & CO3).

Plan for Implementation: Provide more practice on numerical problems and real-time examples

Implementation: Conducted tutorial sessions and problem-solving practice.

Course code	Course Title	Attainment Levels of COs					
CS304PC	Computer Organization and Architecture	**COs**	1	2	3	4	5
		Attained	**1.43**	**1.23**	**1.53**	**1.62**	**1.56**

Gap Analysis:

Students have low understanding of core concepts and problem-solving in computer architecture.

Plan for Implementation: Provide concept explanations with examples and practice problems.

Implementation: Conducted **tutorial sessions, assignments, and problem-solving practice**.

Course code	Course Title	Attainment Levels of COs					
MB401HS	Business Economics & Financial Analysis	**COs**	1	2	3	4	5
		Attained	**2.94**	**1.63**	**2.94**	**2.01**	**2.92**

Gap Analysis:

Most COs are good; students have difficulty in financial analysis concepts (CO2 & CO4).

Plan for Implementation: Provide more numerical problems and real-life financial examples.

Implementation: Conducted **problem-solving sessions and assignments**.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS402PC	Discrete Mathematics	**Attained**	**2.92**	**2.54**	**2.07**	**2.61**	**2.31**

Gap Analysis: CO3 and CO5 show comparatively lower attainment, indicating a need to improve understanding of key concepts and problem-solving skills..

Plan for Implementation: Provide focused practice on weak areas through additional exercises, concept clarification sessions, and use of visual/ICT tools.

Implementation: Conduct remedial classes, regular quizzes, and interactive problem-solving sessions to improve performance in CO3 and CO5.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS404PC	Database Management Systems	**Attained**	**2.30**	**1.64**	**1.86**	**1.50**	**2.12**

Gap Analysis:

Students have low understanding in SQL queries and database design concepts.

Plan for Implementation: Provide more practice on SQL queries and real-time examples.

Implementation: Conducted tutorial sessions, assignments, and query practice.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS507PC	Operating Systems and Computer Networks Lab	**Attained**	**1.94**	**1.94**	**2.05**	**2.16**	**2.16**

Gap Analysis:

Students have low to moderate understanding of OS and networking concepts in practical sessions.

Plan for Implementation: Provide more hands-on experiments and step-by-step lab guidance.

Implementation: Conducted additional lab exercises and practice sessions.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS603PC	Object Oriented Analysis and Design	**Attained**	**2.58**	**2.01**	**3.00**	**2.58**	**2.58**

Gap Analysis: Most COs are achieved; slight difficulty in understanding some design concepts (CO2)..

Plan for Implementation: Provide more case studies and design examples.

Implementation: Conducted **case study discussions and assignments**.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS701PC	Management Science	**Attained**	**2.12**	**2.12**	**1.70**	**1.64**	**2.03**

Gap Analysis: CO3 (1.70), CO4 (1.64), and CO5 (2.03) did not meet the target value of 2.1. Students showed difficulty in applying conceptual knowledge, analytical skills, and decision-making techniques in management-related problems..

Plan for Implementation: Case study discussions

Implementation: Given practice tasks

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS731PE	Multimedia Computing	**Attained**	**1.55**	**2.31**	**1.85**	**1.64**	**2.03**

Gap Analysis:

Students have low to moderate understanding of multimedia concepts and tools.

Plan for Implementation: Provide concept explanation with examples and hands-on practice

Implementation: Conducted tutorial sessions, assignments, and practical exercises.

Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS732PE	BigData Analytics	**Attained**	**2.04**	**2.54**	**2.54**	**2.06**	**2.03**

Gap Analysis:

Moderate understanding; some students face difficulty in data analysis and tools

Plan for Implementation: Provide hands-on practice with tools and real-time datasets.

Implementation: Conducted practical sessions and assignments.

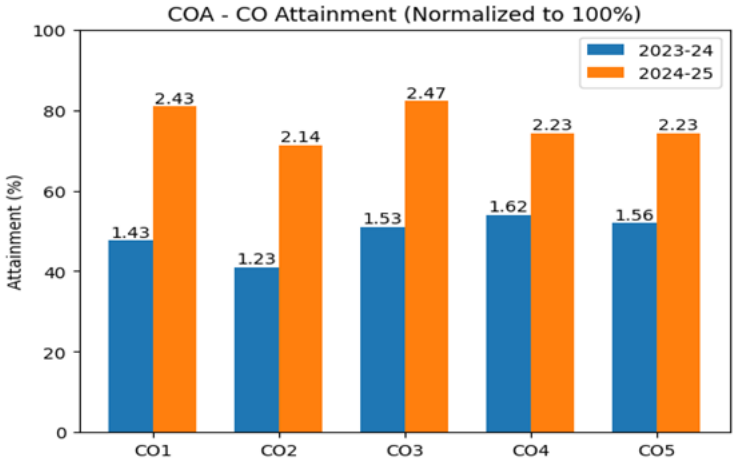
Course code	Course Title	Attainment Levels of COs					
		COs	1	2	3	4	5
CS742PE	Cloud Computing	**Attained**	**1.47**	**1.55**	**1.63**	**1.64**	**1.48**

Gap Analysis: Students have low understanding of cloud concepts and applications.

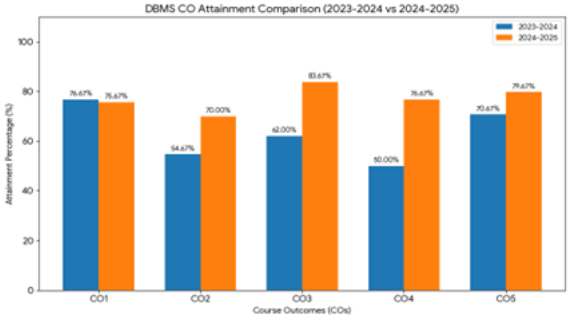
Plan for Implementation: Provide concept explanation with real-time examples and hands-on practice.

Implementation: Conducted group discussions and practical sessions.

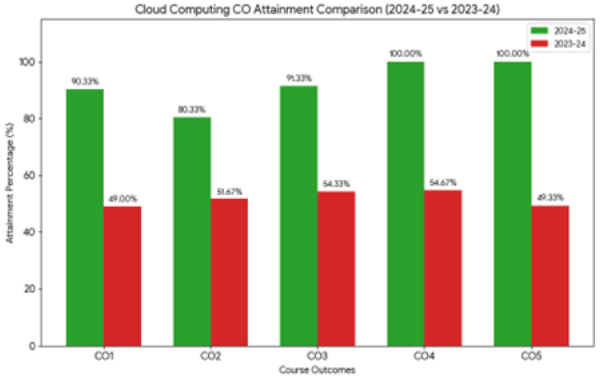
(CS304PC) Computer Organization and Architecture CO's improvement from A.Y 2023-24 to 2024-25:



(CS404PC) Database Management Systems for CO's improvement from A.Y 2023-24 to 2024-25:



(CS742PE) Cloud Computing for CO's improvement from A.Y 2023-24 to 2024-25:



8.1.2 Actions Taken Based on the Results of Evaluation of the POs/PSOs Attainment (20) | Institute Marks : 20.00

A. Documentary Evidence of Identification of Gaps in POs/PSOs Attainment

The Program Outcomes (POs) are adopted in line with the Graduate Attributes prescribed by the National Board of Accreditation (NBA). The Program Specific Outcomes (PSOs) are defined by the department, ensuring alignment with Program Educational Objectives (PEOs), curriculum structure, industry requirements, and societal needs.

Each course in the curriculum has well-defined Course Outcomes (COs), formulated using appropriate action verbs from Bloom's Taxonomy. A systematic CO–PO–PSO mapping process is carried out to establish the relationship between course outcomes and program outcomes. This mapping is performed using a three-point scale:

3 - Strong correlation

2 - Moderate correlation

1 - Weak correlation

The mapping matrix is reviewed and approved by the Board of Studies (BoS) to ensure comprehensive coverage of all POs and PSOs across the curriculum.

Assessment and Evaluation Process

The attainment of POs and PSOs is evaluated using both direct and indirect assessment methods.

Direct Assessment Tools include:

- Continuous Internal Evaluation (CIE)
- End Semester Examinations (SEE)
- Laboratory performance
- Mini and major projects
- Technical seminars (wherever applicable)

Indirect Assessment Tools include:

- Course End Surveys
- Exit Surveys
- Alumni Surveys
- Employer Feedback

As per institutional policy, the weightage assigned is:

- Direct Assessment: 90%
- Indirect Assessment: 10%

Initially, CO attainment is calculated for each course based on student performance in internal and external assessments. Subsequently, PO and PSO attainment levels are computed using the weighted average of CO attainments mapped to the respective PO/PSO.

Computation of PO and PSO Attainment

The final attainment of POs and PSOs is calculated using the formula:

Final Attainment = (Direct Attainment × 0.9) + (Indirect Attainment × 0.1)

The attainment levels are measured on a three-point scale. A target level is predefined for each PO and PSO based on previous academic performance and institutional benchmarks.

If the attained value ≥ target, the outcome is considered attained

If the attained value < target, the outcome is considered not attained

Identification of Gaps and Action Taken

For POs and PSOs that do not meet the target levels, a detailed gap analysis is conducted to identify the root causes. These may include gaps in conceptual understanding, inadequate practical exposure, or lack of industry relevance.

Based on the findings, appropriate corrective measures are implemented, such as:

- Remedial classes for slow learners
- Curriculum enrichment initiatives
- Strengthening project-based learning
- Increased industry interaction and expert lectures
- Faculty Development Programs (FDPs)

The effectiveness of these actions is continuously monitored and reviewed in subsequent academic cycles, ensuring a robust mechanism for continuous improvement in teaching–learning processes and outcome attainment.

PO/PSO Attainment Table for Academic year 2024-25 :

PO/PSO Attainment Table for Academic year 2024-25 :

<table>
<tr><th>POs</th><th>Target Level</th><th colspan="2">Attainment Level</th><th>Observation</th></tr>
<tr><td colspan="5">PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.</td></tr>
<tr><td rowspan="2">PO1</td><td rowspan="2">2.1</td><td>2024-25</td><td>2.69</td><td rowspan="2">PO1 is attained, indicating students effectively apply mathematics, science, and engineering fundamentals to solve complex engineering problems.</td></tr>
<tr><td>2023-24</td><td>2.39</td></tr>
<tr><td colspan="5">Action1: Continue emphasizing real-world problem-solving in lectures and labs.
Action2: Introduce mini-projects and case studies to strengthen application of engineering knowledge.
Action3: Conduct periodic quizzes and assignments to assess and reinforce understanding.
Action4: Provide mentorship and guided learning sessions for students needing additional support.</td></tr>
<tr><td colspan="5">PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.</td></tr>
<tr><td rowspan="2">PO2</td><td rowspan="2">2.1</td><td>2024-25</td><td>2.69</td><td rowspan="2">PO2 is attained, indicating that students are able to identify, formulate, review literature, and analyze complex engineering problems using first principles of mathematics, natural sciences, and engineering sciences.</td></tr>
<tr><td>2023-24</td><td>2.39</td></tr>
</table>

Action 1: Include problem-based learning sessions with real-world case studies.
Action 2: Assign literature review and research tasks to develop analytical skills.
Action 3: Conduct workshops on modeling, simulation, and solution analysis for complex problems

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO3	2.1	2024-25	2.69	PO3 is attained, indicating that students are able to design solutions for complex engineering problems, including system components or processes, while considering public health, safety, and societal, cultural, and environmental aspects.
		2023-24	2.41	

Action1: Conduct design-oriented projects and case studies integrating real-world constraints.
Action2: Organize workshops on sustainable and socially responsible engineering design.
Action3: Encourage team-based assignments to develop practical solution design and decision-making skills.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO4	2.1	2024-25	2.66	PO4 is attained, indicating that students are able to use research-based knowledge and methods, including experimental design, data analysis, and synthesis, to draw valid conclusions for complex problems.
		2023-24	2.40	

Action1: Include laboratory experiments and research projects to enhance investigative skills.
Action2: Conduct data analysis workshops using statistical and computational tools.
Action3: Encourage critical evaluation of research literature and interpretation of results.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO5	2.1	2024-25	2.68	PO5 is attained, indicating that students are able to select and apply appropriate modern engineering and IT tools, including prediction and modeling techniques, while understanding their limitations in complex engineering activities.
		2023-24	2.43	

Action1: Conduct hands-on sessions using industry-standard software and simulation tools.
Action2: Assign projects requiring application of modern tools for problem-solving and modeling.
Action3: Organize workshops and tutorials on emerging engineering and IT tools.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment

PO6	2.1	2024-25	2.69	PO6 is attained, indicating that students are able to apply reasoning and contextual knowledge to assess societal, health, safety, legal, and cultural issues, along with the associated professional responsibilities.
		2023-24	2.51	

Action1: Include case studies highlighting societal and ethical impacts of engineering solutions.
Action2: Organize seminars on legal, health, and safety standards in engineering practice.
Action3: Encourage community engagement and socially relevant projects.
Action4: Conduct workshops on professional ethics and responsibility in engineering.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO7	2.1	2024-25	2.58	PO7 is attained, indicating that students are able to apply ethical principles and adhere to professional ethics, responsibilities, and norms in engineering practice.
		2023-24	2.45	

Action1: Conduct sessions on engineering ethics and case studies of ethical dilemmas.
Action2: Include discussions on professional responsibilities and code of conduct in coursework.
Action3: Encourage participation in seminars and workshops focused on ethical decision-making in engineering.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO8	2.1	2024-25	2.68	PO8 is attained, indicating that students can function effectively as individuals and as members or leaders in diverse, multidisciplinary teams.
		2023-24	2.55	

Action1: Assign group projects and collaborative tasks to strengthen teamwork skills.
Action2: Conduct team-based workshops and competitions to develop leadership and coordination.

Action3: Encourage peer assessments and reflective practices to enhance team performance.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO9	2.1	2024-25	2.68	PO9 is attained, indicating that students are able to communicate effectively on complex engineering activities through reports, presentations, and clear instructions within the engineering community and society.
		2023-24	2.55	

Action1: Conduct workshops on technical writing and report preparation.
Action2: Organize presentation and communication skill sessions.
Action3: Encourage peer reviews and feedback to improve clarity and effectiveness in communication.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO10	2.1	2024-25	2.69	PO10 is attained, indicating that students are able to apply engineering and management principles to manage projects effectively as team members or leaders in multidisciplinary environments.
		2023-24	2.51	

Action1: Include project-based learning with defined management roles for students.
Action2: Conduct workshops on project planning, scheduling, and resource management.
Action3: Encourage use of project management tools and software for real-time tracking and reporting.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PO11	2.1	2024-25	2.64	PO11 is attained, indicating that students recognize the need for and demonstrate the ability to engage in independent and life-long learning in the context of ongoing technological change.
		2023-24	2.39	

Action1: Encourage self-directed learning through online courses and certifications.
Action2: Promote participation in workshops, seminars, and webinars on emerging technologies.

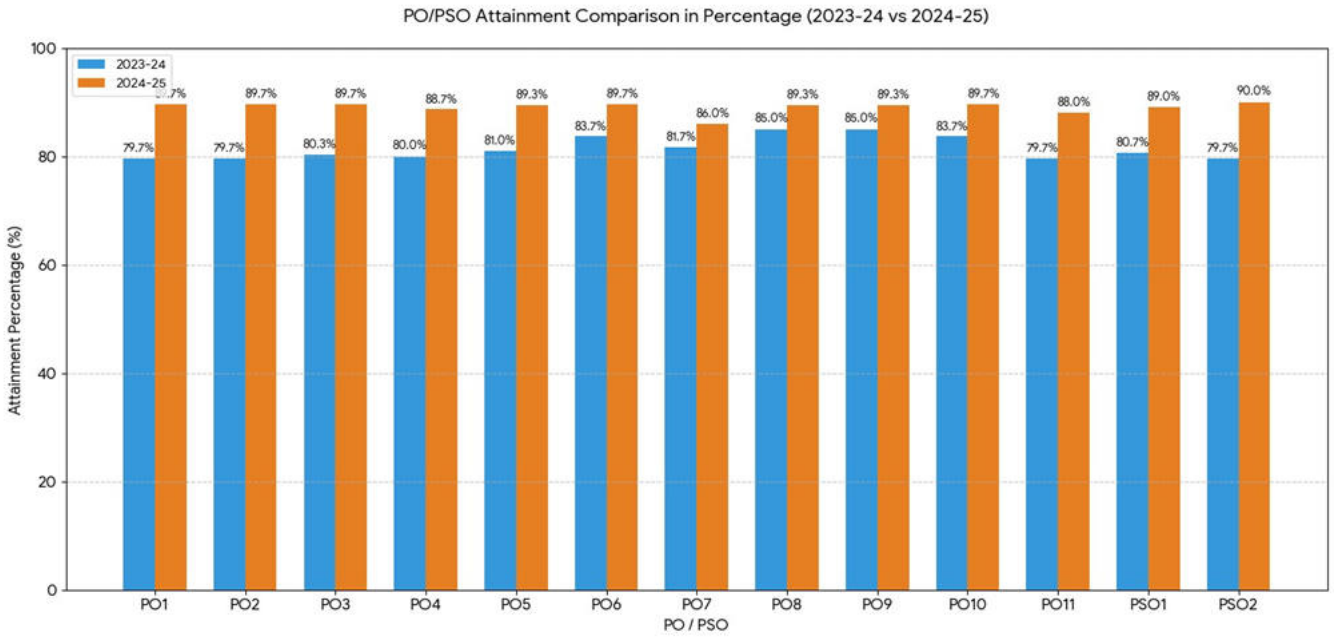
Action3: Assign research-based projects to develop continuous learning and problem-solving skills.

PSOs	Target Level	Attainment Level		Observation
PSO1: Apply mathematical abstraction, algorithms, and data structures to solve complex computing problems using multiple programming paradigms.				
PSO1	2.1	2024-25	2.67	Target Achieved
		2023-24	2.42	
Action 1: Target will be retained and will be observed for next academic year. Action2: To strengthen the domain knowledge and make them job ready graduates, department is planning to introduce more number of professional elective courses.				
PSO2: Design software solutions for multidisciplinary domains using software engineering principles, modern tools, project management, professional ethics, societal responsiveness, and sustainability—enabling career, startup, and higher studies success.				
PSO2	2.1	2024-25	2.7	Target Achieved
		2023-24	2.39	
Action 1: To strengthen the software reliability, network security and emerging technologies innovative techniques are using by Hackathon programs.				

Improvement in PO/PSO and PSO for 2023-24 and 2024-25 Academic Years:

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
2023-24	2.39	2.39	2.41	2.40	2.43	2.51	2.45	2.55	2.55	2.51	2.39	2.42	2.39
2024-25	2.69	2.69	2.69	2.66	2.68	2.69	2.58	2.68	2.68	2.69	2.64	2.67	2.70

Comparison of PO and PSO Attainment Levels for Academic Years 2023–24 and 2024–25



8.2 Academic Audit and actions taken thereof during the period of Assessment (15)

Total Marks 1

Institute Marks : 15

The primary objective of an academic audit is to ensure that departments have established robust and effective quality assurance mechanisms. These mechanisms should encompass well-defined strategies and procedures that are not o applicable but also capable of delivering high-quality inputs and outputs. The audit also evaluates the department's responsiveness to continuous improvement and assesses the adequacy, utilization, and future needs of resources to supp quality education.

A. Availability of External Academic Audit Process

The academic audit process begins with the formation of an audit committee to assess the effectiveness of teaching, learning, and academic operations. The department prepares and presents all necessary documents.

- The department forms an academic audit committee to evaluate teaching and learning practices.
- The department prepares all required documents and evidence before the audit.
- The audit committee visits the department and verifies records, files, and academic activities.
- Faculty members present academic processes, CO-PO attainment, and student performance data.
- The committee interacts with faculty, students, and stakeholders to understand implementation.
- Observations, strengths, and areas for improvement are recorded by the audit team.
- A formal audit report is submitted to the Head of Department and Principal.
- The department then prepares an Action Taken Report (ATR) for each observation.
- Corrective actions are implemented within the given timeline.
- The ATR is submitted to IQAC and monitored for compliance and continuous improvement.

Frequency of Audit:

- According to the organization's guidelines, internal audits are conducted **twice in every year** after completion of each semester.
- External audits are conducted **once in a year**.
- This review ensures independent evaluation of processes and performance.
- Additional audits may be scheduled if the organization identifies a need for further verification.
- Audit is done by Members appointed by HoD-Chairman of Audit Committee along with External Member selected from other colleges/Universities and from Industry.

Step-by-Step Process

1. Notification and Audit Planning

- The department appoints internal auditors from senior faculty and IQAC members.
- External auditors are invited from reputed institutions or universities to provide an independent evaluation.
- Audit schedule is circulated in the department and to Audit members.
- Audit evaluation parameters and checklist are shared beforehand.
- Departments are instructed to prepare evidence and documentation.

2. Academic Audit Committee Formation

- The Academic Audit Committee is formed to ensure systematic evaluation and continuous improvement of academic processes, comprising internal auditors (HOD, senior faculty, IQAC) for reviewing course files, assessments, and teaching practices, and external auditors (subject experts, industry expert, IQAC nominee, Principal) for independent validation and benchmarking of curriculum, CO–PO alignment, and industry connect. This dual committee structure ensures objectivity, accountability, and effective implementation of audit recommendations.

3. Composition of the Academic Audit Committee

- The composition of the Academic Audit Committee is determined by the Department Academic Committee (DAC).
- The Internal Academic Audit is conducted by a committee comprising senior faculty members, Heads of Departments (HoDs), and academic coordinators. The committee provides observations, recommendations, and corrective measures to ensure continuous improvement in academic quality, alignment with outcome-based education (OBE), and adherence to institutional policies.

S. No	Role	Responsibility
1	Chairperson – HOD	Leads audit, assigns tasks, finalizes report
2	Senior Faculty Members	Review course files, lab manuals, assessments, and teaching-learning processes
3	IQAC Coordinator	Maintains records, coordinates with departments, and ensures compliance with quality standards
4	Department Faculty Members	Provide documentation, clarify audit observations, and assist in evidence preparation
5	Principal	Approves committee and ensures institutional support

Composition of the External Academic Audit

An External Academic Audit is an independent review by external experts to assess the quality and effectiveness of academic processes. It evaluates curriculum implementation, teaching-learning methods, assessment practices, and CO–PO–PSO attainment. The audit ensures compliance with regulatory standards and identifies strengths, gaps, and improvement actions. Its outcomes support continuous improvement.

Table No. 8.2.1.b – Composition of the External Academic Audit

S. No	Name	Designation	Organization	Role
1				Chairman
2				Member
3				Member
4				Industry Expert

B. Plan of action to address the recommendations

Based on the recommendations of the External Academic Audit, a structured action plan is prepared by the department. The academic audit evaluation was initiated to assess the effectiveness of the teaching-learning processes, CO/PO attainment, and overall academic performance. The process involved reviewing course documentation, student performance records, lab and project work, and assessment outcomes. Faculty meetings and discussions were conducted to identify gaps in course delivery, integration of digital tools, and alignment of assessments with defined outcomes. Preliminary observations were made to highlight areas needing improvement and to plan remedial actions for low-attainment courses, ensuring timely and structured monitoring before the formal evaluation.

Table No: 8.2.2 – Evaluation Parameters for Audit

S. No	Audit Criteria	Evaluation Parameters
1	Department Academic Calendars	• Prepared as per the institution's schedule; • includes working days, events, and exams; • communicated to stakeholders
2	Department Time Tables (Class-wise & Faculty-wise)	• Timely preparation; • class-faculty mapping; • no clash in schedules
3	Faculty Workload	• Workload as per norms (AICTE/JNTUH); • Balanced teaching, research, and admin load.
4	List of Courses	• Updated as per R18/R22 regulations; • includes electives and open electives.
5	Subject Allotments	• Based on specialization & experience; • Workload justification.
6	Lesson Plan Approval Copies	• Prepared before semester start; • Contains CO mapping, activities, and references.

7	Course Files	• Contains syllabus, notes, assignments, and CO-PO mapping; • updated regularly.
8	Modern Pedagogical Teaching Methods	• Use of ICT tools; active learning strategies; • Evidence of implementation
9	Faculty Feedback Analysis	• Feedback collected online/offline; • Action taken recorded.
10	Mentoring & Counselling	• Mentor-mentee ratio; • Regular interaction records
11	Guest Lectures	• Number conducted vs planned • Attendance and Feedback • Industry/Expert relevance
12	Identification of Slow Learners	• Method used (results, faculty input); • List maintained.
13	Motivation to Advanced Learners	• Support for projects/certifications; • Participation in competitions.
14	Student Internships	• Number of students placed in internships; • Duration & relevance.
15	Student Workshops	• Skill-based workshop frequency; • Industry relevance.
16	Industrial Visits	• Visit relevance to syllabus gaps; • Attendance and report evidence.
17	Student Hackathons	• Participation/hosting records; • Awards and outcomes.
18	Remedial Classes	• Schedule and attendance; • Improvement tracking.
19	Attainment of Course Outcomes	• CO attainment calculation method; • Direct & indirect methods used.
20	Attainment of PO & PSO	• PO-PSO attainment matrix; • Targets vs achieved.
21	Mid Exams Verification	• Blueprint matching;

		• CO mapping present.
22	Question Papers (BTL Level Compliance)	• CO mapped; • As per Bloom's taxonomy
23	Result Analysis & Action Plan	• Pass %, toppers, failures; • Corrective actions prepared.
24	Curriculum Gap Identification	• Gap analysis report; • Action through value-added courses.
25	Course Outcomes Listed	• COs defined, mapped, and displayed.
26	PEOs	• PEOs defined & validated in BOS/Advisory meetings
27	POs & PSOs	• As per NBA format; • Displayed and mapped.
28	DAC & PAC Action Taken Reports	• Meetings conducted; • Action items recorded
29	Placement Assistance	• Number of drives/companies; • Soft-skill training initiatives.
30	IQAC Review	• Timely IQAC review reports; • Implementation of recommendations.

Action Taken by Department

After the completion of the Academic Audit, the department prepares an Action Taken Report (ATR) addressing each observation and recommendation made by the audit committee.

The Action Taken Report clearly outlines:

- Action Planned / Implemented
- Responsible Faculty / Committee
- Timeline for Completion
- Status of Compliance

The report ensures that corrective measures are initiated within the stipulated time and continuous improvement is maintained in all academic processes.

Department Initiatives

The department organizes various **technical workshops, training programs, and cultural activities**, including AWS workshops, Blockchain training, and Full Stack Development sessions. These initiatives strengthen students' practical skills and align with Outcome-Based Education (OBE) practices.

Strengths

- Adequate faculty strength with good SFR

- Active research culture with publications and patents
- Strong faculty development initiatives
- Good placement record and student success rate
- Emphasis on skill development and value-added courses

Areas for Improvement

- Increase number of faculty with PhDs
- Enhance funded research and consultancy projects
- Improve industry collaborations and MoUs
- Strengthen PG program admissions and outcomes

Conclusion

The Department of Computer Science and Engineering demonstrates strong academic performance, active research involvement, and effective student engagement practices. The department is steadily progressing towards meeting **NBA accreditation criteria**, particularly in areas of **OBE implementation, research, and continuous improvement**, while focusing on further enhancements in research funding and industry interaction.

The final Action Taken Report is formally submitted to the Internal Quality Assurance Cell (IQAC) and the principal for review, approval, and further monitoring.

DEPARTMENT ACADEMIC AND ADMINISTRATIVE AUDIT

EXTERNAL ACADEMIC AND ADMINISTRATIVE AUDIT REPORT

(ACADEMIC YEAR: 2024-25)

Date of Audit: 12-09-2026

Name and Designation of Audit Member-1: **Dr. Dr. M G Sethuraman**,

Emeritus Professor, Gandhigram Rural Institute,

Deemed to be University, Gandhigram.

Name and Designation of Audit Member-2: **Dr. G. Charles Babu, Professor**,

Department of CSE, Gokaraju Rangaraju Institute of Engineering and Technology, Bachupally, Hyderabad.

1. **Name of the Department to be Audited: Computer Science and Engineering**
2. **Programs Offered: UG - B. Tech (CSE)**

	UG	**PG**
Name of the Program	B. Tech	M. Tech
Year of establishment	2001	2007
Intake	180	18

3. **No. of Faculty & Students:**

Designation	UG		PG	
	Sanctioned	Actual	Sanctioned	Actual
Professor	3	3	0	0
Associate Professor	6	4	1	1
Assistant Professor	18	29	2	2

	Year	No. of Boys	No. of Girls
No. of Students (UG)	II	97	90
	III	87	97
	IV	110	94
	Total	294	281
No. of Students (PG)	I	-	-
	II	-	-
	Total	-	-

4. **Faculty Information and Contributions**

Name of the Program(s):

Total Faculty	
Required Faculty	30
Available Faculty	39

No. of PhDs	8

Total Number of Faculty with PhD in the Department*100	
% of PhD Holders =	=20.5
Total Number of Faculty Available in the Department	
Total Number of Students (UG) in the Department	
Student Faculty Ratio(SFR) for UG =	=14.7
Total Number of Faculty Available in the Department	
Total Number of Students (PG) in the Department	
Student Faculty Ratio(SFR) for PG =	= Nil
Total Number of Faculty Available in the Department	

Faculty Publications

A	B	C	D	E	F	G	H	I	J
S. No	**Name of the Faculty**	**No. of Papers Published**			**Papers Published in Conferences**	**Papers Published in Journals**	**Total Papers Published H=F+G**	**H-Index (Till Date)**	**No. of Citations (Till date)**
		Indexed in SCIE/ ESCI	**Indexed in Scopus**	**Indexed in Web of Science**					
1	Dr. C. H. N. Santhosh Kumar		2	2	1	2	3	8	213

2	Mr. CH. Krishna Prasad		1		1		1	1	4
3	Mr. G.L.N.V Kartheek		1					11	
4	Mr.V.Achiredy		1						
5	Dr.Y.V.R Naga Pawan,		1	1	1	0	1	8	355
6	Mr.V. Jeevan Babu		1						
7	Dr.M.Murugesan			1				4	26
8	Dr. M.V. Siva Prasad				1			20	1384

Book Chapters Published by Faculty

S. No.	Name of the Faculty	Book Chapter Title	Title of the Book	Name of the Publisher with ISBN	Indexed In
1	Dr.Y.V.R Naga Pawan	Enhancing Connectivity And Data-Driven Decision-Making For Smart Agriculture By Embracing 6g Technology	Development of 6G Networks and Technology	Wiley-9781394230655	Scopus

Faculty Intellectual Property Rights (IPR)

S. No.	Name of the Faculty	Title of the Patent	File Number	Status (Published/

				Granted/ Design)
1	Mr. V. Jeevanbabu	Next-Gen Agriculture: Iot, Cloud Computing, And Ai For Detecting Plant Diseases	202441058780 A	09/08/2024
2	Mr.V. Achireddy	Integration Of The Internet Of Things And Ai For Monitoring Climate, Earthquakes, And Air Quality	202411057883 A	16/08/2024
3	Mr.V. Achireddy	Intelligent Path Travel Cost Estimation In Large-Scale Networks Using Advanced Machine Learning	202441064131 A	30/08/2024
4	Dr. M. Murugesan	Design Of Intelligent Machine Learning Based Chip Control Oxygen Concentrator With Filter	202441079806 A	25/10/2024
5	Mrs. M. Anusha	Machine Learning-Based Iot Healthcare Solution For Personalized Wellness Recommendations	202441086544 A	15/11/2024
6	Mrs. J. Vijayasree	Design Augmented Intelligence Based Skin Cancer Classification And Prediction Framework Analysis	202441087380 A	22/11/2024
7	Mrs. P. Hema Sai	Adaptive System For Real-Time Noise Reduction In Web Data Using Dynamic User Interest Profiling	202441087502 A	22/11/2024
8	Mrs. G. Prashanthi	Identification Of Fungal Disease Of Nutmeg Leaves Using Machine Learning Approach	202441088321 A	22/11/2024
9	Mr.V. Achireddy	Artificial Intelligence-Based System For Adaptive User Interface Customization	202441088831 A	22/11/2024
10	Mrs. A.Srilatha	Automated Code Synthesis Using Deep Learning Techniques	202441088833 A	22/11/2024

11	Mr. B. Nageswar Rao	Social Media Management Tool For Brand Reputation Management	202441091563 A	29/11/2024
12	Mr. Mohammed Abdul Gafar	Secure Data Storage System Using Blockchain Technology	202441091566 A	29/11/2024
13	Mr. B. Govinda Rao	Decentralized Insurance Claim Processing Using Blockchain Technology	202441091975 A	06/12/2024
14	Mr. G.L.N.V Kartheek	Blockchain Based Secure Data Sharing Method For Distributed Systems	202441093777 A	06/12/2024
15	Mrs.Ch. Nirmala	Automated News Verification And Summarization System For Accurate Information Dissemination	202441094511 A	13/12/2024
16	Mrs.Ch. Mamatha	System And Method For Sentiment Analysis Of Online Product Reviews Using Opinion And Text Mining	202441094512 A	13/12/2024
17	Mrs. V. Jyothi	Method For Blockchain-Based Academic Credential Verification	202441095933 A	13/12/2024
18	Dr. K. Srinivasa Reddy	Block Chain Enabled Secure Document Management System	202441096896 A	13/12/2024
19	Mr. K. Bikshapati	Block Chain Technology For Enhanced Cyber security In Networks	202441099036 A	20/12/2024
20	Mrs. K. Anusha	An Iot-Based Industrial Cloud System For Real-Time Wastewater Monitoring And Control	202441098176 A	20/12/2024

Faculty Sponsored Projects

S. No.	Name of the Faculty (PI/Co-PI)	Title of the Project	Funding Agency	Project Duration	Sanctioned Amount (Lakhs)	Amount Received Lakhs)
-	-	-	-	-	-	-

Faculty Consultancy Projects

S. No.	Name of the Faculty	Title of the Project	Industry/ Organization	Consultancy Project (Duration)	Sanctioned Amount (Lakhs)	Amount Received (Lakhs)
	-	-	-	-	-	-

Faculty Participation in FDPs/Workshops/STTPs/etc. (Recognized by AICTE /UGC/ DST / NITTTR/NPTEL)

S.No	Name Of The Faculty	Name Of The Fdp /Workshop /Sttp	Date(S) Of The Program	Duration	Organized Institute	Online/ Offline
1	Dr Y V R Naga Pawan	Fundamental Data Structures	01.01.2024	0	ACM	Online
2	Mrs. J Vijayasree	Programming In Java	01.07.2025 To 01.10.2025	84	NPTEL	Online
3	Mrs. N Mownika	Software Engineering	01.07.2024 To 01.10.2024	84	NPTEL-AICTE	Online
4	Mrs. N Mownika	Software Engineering	01.07.2024 To 01.10.2024	84	NPTEL-AICTE	Online
5	Dr Kothapalli Mandakini	Fundamentals Of Artificial Intelligence	01.07.2024 To 02.11.2024	84	IIT Kharaghpur	Anurag Engineering Colleg
6	Mr. Ch Krishna Prasad	Nep Orientation And Sensitization Programme	02.06.2025 To 13.06.2025	8	UGC-MMTTC University Of Kerala	Online
7	Mr. Ch. Venkata Navi	Quantum Computing Impact On Cryptography And Cyber Security In Defense	02.12.2024 To 07.12.2024	6	Sasurie College Of Engineering	Online

8	Mr. Md Abdul Gafar	ATAL	02.12.2024 To 07.12.2024	7	Sasurie College Of Engineering(Atal)	Online
9	Mr.Ch Krishna Prasad	Data Science Mastry	05.02.2024 To 10.02.2024	6	VNRVJIT	Online
10	Mr.Ch Krishna Prasad	Data Science Mastery	05.02.2024 To 10.02.2024	6	VNRVJIT	Online
11	Mr.Ch Krishna Prasad	Problem Solving Through Programming In C	05.07.2024 To 05.10.2024	72	NPTEL	Online
12	Mrs. S Sowjanya	Introduction To Internet Of Things	05.07.2025 To 06.10.2025	12	SWAYAM	Online
13	Mr. B Govindarao	Quantum Computing Impact On Cryptography And Cyber Security In Defense	05.12.2024 To 07.12.2024	06	Sasurie College Of Engineering	Online
14	Mr.P. Kishore	Programming In Java	06.02.2025 To 17.04.2025	94	NPTEL	Online
15	Dr Y V R Naga Pawan	Applied Cloud Computing For Full Stack Web Development	08.01.2024 To 12.01.2024	5	ATAL AICTE MICROSOFT SAP EDUNET	Online
16	Mrs. N Mownika	Applied Cloud Computing For Full Stack Web Devvelopment	08.01.2024 To 12.01.2024	5	MICROSOFT,SAP & AICTE	Online
17	Mrs. N Mownika	Blockchain Technology	08.01.2024 To 12.01.2024	5	D. Y Patil Deemed To Be University	Online
18	Mrs. Anusha Merugu	Applied Cloud Computing For Full Stack Web Development	08.01.2024 To 12.01.2024	5	Microsoft,Sap And AICTE	Online
19	Mr. Ch Krishna Prasad	Black Chain Technology	08.01.2024 To 12.01.2024	5	Excel R	Online
20	Mr. Ch Krishna Prasad	Black Chain Technology	08.01.2024 To 12.01.2024	4	Excelr Edtech Private Limited	Online
21	Mr. Ch Krishna Prasad	The Art Of Writing Research Article And Publishnig In High Impact Scopus Sci E Journals	09.01.2024	1	MLRIT	Online
22	Mr. Ch Krishna Prasad	The Art Of Writing Research Article And Publishing In High Impact Scopus Or Scie Journals	09.06.2024	1	MLRIT	Online
23	Mr. Ch Krishna Prasad	Problem Solving Through Programming In C	10.07.2024 To 16.10.2024	84	NPTEL AICTE	Online
24	Dr Y V R Naga Pawan	Introduction To Pythons Unittest Framework	11.10.2024	0	ACM	Online
25	Dr Y V R Naga Pawan	Advanced Fucntionality Using Python Classes	11.10.2024	0	ACM	Online
26	Mr. K. Veeranjaneyulu	Next Generation Concrete Technology Utilizing Nanomaterials For The Development Of Strong And Smart Infrastructure	14.07.2025 To 19.07.2025	06	Anurag Engineering College	Ananthagiri
27	Dr Y V R Naga Pawan	Quantum Computing Alogorithms And Applications	15.07.2024 To 19.07.2024	5	NIT Warangal	Online
28	Mrs. Pagadala Janaki	Introduction To Internet Of Things	15.07.2025 To 24.10.2025	12	NPTEL	Online

29	Mr. Chejarla Venkata Navi	Orientation And Sensitization Programme	15.12.2025 To 23.12.2025	8	UGC MMTTC, JNTU Hyderabad	Online
30	Mr. K Bikshapati	ATAL Fdp	16.12.2024 To 21.12.2024	6	Gopal Narayan Singh University	Online
31	Dr Y V R Naga Pawan	Applied AI	18.11.2024 To 22.11.2024	5	Microsoft Edunet Sap AICTE	Online
32	Mr. Ch Krishna Prasad	Atal Fdp On Artificial Intelligence For Geospatial Data Analysis	18.11.2024 To 23.11.2024	6	VJIT	Online
33	Mrs. M Sadalaxmi	Data Analytics With Python	19.01.2025 To 20.04.2025	12	NPTEL	SWAYAM
34	Dr Y V R Naga Pawan	Data Preprocessing	19.05.2024	0	ACM	Online
35	Dr Y V R Naga Pawan	Python Apprentice	19.05.2024	0	ACM	Online
36	Mr. Ch Krishna Prasad	Data Analyst	19.06.2024 To 23.06.2024	5	APSSDC	Online
37	Mr.Ch Krishna Prasad	Data Analyst	19.06.2024 To 23.06.2024	5	APSSDC	Online
38	Mr. K. Veeranjaneyulu	Nep Orientation And Sensitization Programme	19.06.2025 To 27.06.2025	06	UGC MMTTC, Hyd	Online
39	Dr T V Suresh Kumar	Regional Research Symposim On Project Based Learning	19.12.2024 To 21.12.2024	3	Anurag University	Offline
40	Mr. Ch Krishna Prasad	Rrspbl	19.12.2024 To 21.12.2024	3	IUCEE	Anurag University Hyderab
41	Mr. Ch Krishna Prasad	Mern Stack Development	20.01.2025 To 24.01.2025	5	AICTE & Tech Saksham	Online
42	Dr Kothapalli Mandakini	Techsaksham	20.01.2025 To 24.01.2025	5	Microsoft Edunet	Anurag Engineering Colleg
43	Mrs. A Srilatha	Data Analyticcs With Python	21.01.2024 To 16.04.2024	84	NPTEL	Online
44	Dr Kothapalli Mandakini	Mmttp	21.07.2024 To 02.08.2024	11	Indian School Of Mines	Anurag Engineering Colleg
45	Mr.Vummadi Achireddy	Data Analytics With Python	22.01.2024 To 12.04.2024	84	NPTEL	Online
46	Mr.Vummadi Achireddy	Data Analytics With Python	22.01.2024 To 12.04.2024	84	NPTEL	Online
47	Mrs. Anusha Merugu	Machine Learning	22.01.2024 To 15.03.2024	56	NPTEL	Online
48	Mrs. A Srilatha	Data Analytics With Python	22.01.2024 To 16.04.2024	56	NPTEL	Online
49	Mrs. Pagadala Janaki	Cloud Computing	22.01.2025 To 19.04.2025	12	NPTEL	Hyderabad

50	Mrs. J Vijayasree	Data Analytics With Python	22.01.2025 To 19.04.2025	12	NPTEL	Online
51	Mr. Ch Krishna Prasad	Data Science Summit	22.02.2025	1	Anurag University	Anurag University Hyderab
52	Mr. Vummadi Achireddy	Machine Learning And Deep Learning Fundementals	22.07.2024 To 11.10.2024	84	NPTEL	Online
53	Mr. Vummadi Achireddy	Machine Learning And Deep Learning Fundementals	22.07.2024 To 11.10.2024	84	NPTEL	Online
	Mrs. A Srilatha	Introduction To Machine Learning	22.07.2024 To 11.10.2024	84	NPTEL	Online
55	Mrs. A Srilatha	Machine Learning And Deep Learning Fundementals	22.07.2024 To 11.10.2024	84	NPTEL	Online
56	Mrs. A Srilatha	Machine Learning And Deep Learning Fundementals	22.07.2024 To 11.10.2024	84	NPTEL	Online
57	Mr. Vummadi Achireddy	Introduction To Machine Learning	22.07.2024 To 13.09.2024	56	NPTEL	Online
58	Mr. Vummadi Achireddy	Introduction To Machine Learning	22.07.2024 To 13.09.2024	56	NPTEL	Online
59	Dr Kothapalli Mandakini	Mmtp	22.10.2024 To 31.10.2024	9	JNTUH	Anurag Engineering Colleg
60	Dr Y V R Naga Pawan	Exceptions And Command Line Arguments	23.05.2024	0	ACM	Online
61	Dr Y V R Naga Pawan	File Operations In Python	23.05.2024	0	ACM	Online
62	Dr Y V R Naga Pawan	Modules And Virtual Environments	23.05.2024	0	ACM	Online
63	Dr Y V R Naga Pawan	Creating Classes Handling Errors Importing Modules	23.05.2024	0	ACM	Online
64	Dr Y V R Naga Pawan	Data Analytics With Python	24.01.2024 To 24.04.2024	72	IIT MADRAS	Online
65	Dr Y V R Naga Pawan	Migrating From Python Two To Python Three	24.05.2024	0	ACM	Online
66	Dr Y V R Naga Pawan	Getting Started With Classes In Python	24.05.2024	0	ACM	Online
67	Dr Y V R Naga Pawan	Working With Inheritance In Python	24.05.2024	0	ACM	Online
68	Dr Y V R Naga Pawan	Python For Data Science	24.07.2023 To 24.09.2023	28	SWAYAM IIT MADRAS	Online
69	Mr. Md Abdul Gafar	Data Base Management System	25.01.2025 To 25.03.2025	56	NPTEL-AICTE	Online
70	Mr. Ch Krishna Prasad	Research Methodologies	27.01.2024	01	Anurag Engineering College	Anurag Engineering Colleg

71	Mrs. G. Prashanthi	Theory Of Computation	22.07.2024 13.09.2024	56	NPTEL	Online
72	Mrs. G. Prashanthi	Problem Solving Through Programming In C	10.07.2024 To 16.10.2024	84	NPTEL	Online
73	Mrs. G. Prashanthi	Introduction To Machine Learning	20.01.2025 To 11.04.2025	84	NPTEL	Online

Faculty Certification

S. No.	Name Of The Faculty	Name Of The Course/Certification	Duration	Certifying Agency
1	Mr.Ch. Krishna Prasad	Problem Solving Through Programming In C	12 Weeks	NPTEL
2	Dr Y.V.R Naga Pawan	Data Analytics With Python	12 Weeks	NPTEL
3	Dr Y.V.R Naga Pawan	Python For Data Science	4 Weeks	NPTEL
4	Dr.V.M. Subramanyam	Operating System Fundementals	12 Weeks	NPTEL
5	Dr.V.M. Subramanyam	Cloud Computing	12 Weeks	NPTEL
6	Mrs. M. Anusha	Machine Learning	8 Weeks	NPTEL
7	Mr. G.L.N.V Kartheek	Introduction To Internet Of Things	12 Weeks	NPTEL
8	Mr. K Naresh Kumar	Programming Data Structures And Algorithms Using Python	8 Weeks	NPTEL
9	Mr. N.Siva Kumar	Introduction To Internet Of Things	12 Weeks	NPTEL
10	Dr. M. Murugesan	Big Data Computing	8 Weeks	NPTEL

11	Mr.Ch. Krishna Prasad	Programming In Python	12 Weeks	NPTEL
12	Dr.T. V. Suresh Kumar	Operating System Fundamentals	12 Weeks	NPTEL
13	Mr. V.Jeevanbabu	Introduction To Internet Of Things	12 Weeks	NPTEL
14	Mrs. A. Srilatha	Introduction To Machine Learning	8 Weeks	NPTEL
15	Mrs. A. Srilatha	Data Analytics With Python	12 Weeks	NPTEL
16	Mrs. A. Srilatha	Machine Learning And Deep Learning Fundamentals And Applications	12 Weeks	NPTEL
17	Mr. Md Ayub Khan	Introduction To Internet Of Things	12 Weeks	NPTEL
18	Mrs. B. Swetha	Introduction To Internet Of Things	12 Weeks	NPTEL
19	Mrs. P. Hemasai	Iot	12 Weeks	NPTEL
20	Mrs. P. Bhavani	Iot	12 Weeks	NPTEL
21	Mrs. N.Mownika	Software Engineering	12 Weeks	NPTEL
22	Mr. V.Achireddy	Data Analytics With Python	12 Weeks	NPTEL
23	Mr.V. Achireddy	Introduction To Machine Learning	8 Weeks	NPTEL
24	Mr. V.Achireddy	Machine Learning And Deep Learning Fundamentals	12 Weeks	NPTEL
25	Mr. V.Achireddy	Electrical Machines I	12 Weeks	NPTEL
26	Mr. V.Achireddy	Digital Circuits	12 Weeks	NPTEL
27	Mrs. V. Jyothi	Introduction To Internet Of Things	12 Weeks	NPTEL
28	Mrs. Ch. Mamatha	Internet Of Things	12 Weeks	NPTEL
29	Mrs. J. Vijayasree	Programming In Java	12 Weeks	NPTEL

30	Mrs.Y. Venkatakumari	Introduction To Internet Of Things	12 Weeks	NPTEL
31	Mrs.Ch. Nirmala	Introduction To Internet Ofthings	12 Weeks	NPTEL
32	Mrs.Ch. Mamatha	Internet Of Things	12 Weeks	NPTEL
33	Dr. K. Srinivasa Reddy	Practical Cyber Security For Cyber Security Practitioners	12 Weeks	NPTEL
34	Mrs.G. Prashanthi	Theory Of Computation	8 Weeks	NPTEL
35	Mrs.P. Kishore	Programming In Java	12 Weeks	NPTEL
36	Mrs.G. Prashanthi	Problem Solving Through Programming In C	12 Weeks	NPTEL
37	Dr K. Mandakini	Introduction To Internet Of Things	12 Weeks	NPTEL
38	Mrs.K. Anusha	Introduction To Internet Of Things	12 Weeks	NPTEL
39	Dr. Ch. N. Santhosh Kumar	Introduction Of Internet Of Things	12 Weeks	NPTEL
40	Mrs. G. Prashanthi	Introduction To Machine Learning	12 Weeks	NPTEL
41	Mr. Kbikshapathi	Programming In Java	12 Weeks	NPTEL
42	Dr. J. Srinivas Rao	Foundation Of Cloud Iot Edge Ml	8 Weeks	NPTEL
43	Dr. J. Srinivas Rao	Python For Data Science	4 Weeks	NPTEL
44	Dr. J. Srinivas Rao	The Joy Of Computing Using Python	12 Weeks	NPTEL
45	Dr. J. Srinivas Rao	Introduction To Internet Of Things	12 Weeks	NPTEL
46	Dr. J. Srinivas Rao	J Srinivas Rao	8 Weeks	NPTEL
47	Dr. J. Srinivas Rao	Machine Learning For Engineering And Science Applications	12 Weeks	NPTEL

48	Dr. J. Srinivas Rao	Introduction To Industry And Industrial Internet Of Things	12 Weeks	NPTEL
49	Mrs. P. Janaki	Cloud Computing	12 Weeks	NPTEL
50	Mrs. P. Janaki	Introduction To Internet Of Things	12 Weeks	NPTEL
51	Mr. Ch. Krishna Prasad	Programming In Python	12 Weeks	NPTEL
52	Mr. K.Veeranjaneyulu	Machine Learning	8 Weeks	NPTEL
53	Mr.Ch. Krishna Prasad	Problem Solving Through Programming In C	12 Weeks	NPTEL
54	Mr. Ch. Krishna Prasad	Introduction To Internet Of Things	12 Weeks	NPTEL
55	Mr. B. Govinda Rao	Introduction To Internet Of Things	12 Weeks	NPTEL
56	Mrs. M. Sadalaxmi	Data Analytics With Python	12 Weeks	NPTEL
57	Mrs. M. Anusha	Data Science For Engineers	8 Weeks	NPTEL
58	Mrs. M. Anusha	Data Analytics With Python	12 Weeks	NPTEL
59	Mr. B. Govinda Rao	Introduction To Internet Of Things	12 Weeks	NPTEL
60	Mr. B. Govinda Rao	Cloud Computing	12 Weeks	NPTEL
61	Mrs. V. Ramadevi	Machine Learning	8 Weeks	NPTEL
62	Mrs. V. Ramadevi	Programming In Java	12 Weeks	NPTEL
63	Mrs. S. Sowjanya	Introduction To Internet Of Things	12 Weeks	NPTEL

Interaction with outside world (BOS/ Academic Council/ Guest Lecture/ Session Chair Person/Co-Chair Person)

S.No.	Name of the Faculty	Type of Program	Lecture Topic	Duration	Organization	Place of Program
1	Dr. M.V. Siva Prasad	Session Chair	Conference Session Chair – NCFT2025 on 30th May 2025	2 Hours	Gurunanak University	Hyderabad

5. **Students Enrollment and Progression**

Number of seats filled under various quotas and quality of students No. of Students (UG):

UG	OC		BC		SC		ST		PWD		SPORTS		EWS	
	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
EAMCET Starting Rank	67185	51135	41825	10012	69681	53014	51836	113706	-	-	-	-	-	-
EAMCET Closing Rank	130955	131860	151163	146482	163873	174669	174891	171423	-	-	-	-	-	-

No. of Students (PG):

PG	OC		BC		SC		ST		PWD		SPORTS		EWS	
	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
Gate Starting Rank	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Gate Closing Rank	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PGCET Starting Rank	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PGCET Closing Rank	-	-	-	-	-	-	-	-	-	-	-	-	-	-

List of Detained Students

Detained Due to Credit Shortage:

Sl. No.	Hall Ticket Number	Student Name	Year& Semester	Required Credits	Credits Earned	Credit Shortage	Reason for Detention
1	22C11A05A3	Ch. Pravallika	II-II	20	17.5	2.5	Due to shortage of credits

Detained Due to Attendance Shortage:

S.No.	Hall Ticket Number	Student Name	Year& Semester	Total Working Days	Days Attended	Attendance %	Required %	Reason for Detention
1	23C11A0525	Aravind Mudusu	II-I	527	0	0	75	Shortage of Attendance
2	23C11A0530	Bhagyasri Bari	II-I	527	0	0	75	Shortage of Attendance
3	23C11A0533	Bharath Kumar Bhukya	II-I	527	193	37	75	Shortage of Attendance

4	23C11A0539	Divyasri Banoth	II-I	527	183	35	75	Shortage of Attendance
5	23C11A05A9	Praneeth Sripathi	II-I	525	0	0	75	Shortage of Attendance
6	23C11A05B0	Prasanna Chilaka	II-I	525	0	0	75	Shortage of Attendance
7	23C11A05H6	Yogamrutha Oggu	II-I	502	0	0	75	Shortage of Attendance
8	24C15A0502	Dinesh Dasari	II-I	502	121	24	75	Shortage of Attendance
9	22C11A0508	Ajayendra Varma Bekkam	III-I	536	126	24	75	Shortage of Attendance
10	22C11A0544	Hazira Yasmin Shaik	III-I	536	14	3	75	Shortage of Attendance
11	22C11A05C3	Saicharan Bittu	III-I	523	0	0	75	Shortage of Attendance
12	23C15A0510	Naveen Kanchi	III-I	522	0	0	75	Shortage of Attendance
13	21C11A05B1	Rajesh Narra	IV-I	447	6	1	75	Shortage of Attendance

14	23C11A05D4	Narapongu Siri	II-II	403	0	0	75	Shortage of Attendance
15	23C11A0586	Shaik Naseeruddin	III-II	448	0	0	75	Shortage of Attendance
16	23C11A05B8	Gugulothu Sai	III-II	448	82	18	75	Shortage of Attendance
17	20C11A05E9	Suman Marisetty	IV-II	317	54	17	75	Shortage of Attendance
18	21C11A0560	Laxmi Prasanna. M	IV-II	317	108	34	75	Shortage of Attendance

Performance of Graduating Students (Batch Wise)

No.of students admitted in corresponding first year (UG) and students admitted in second year through lateral entry (m) = 212

No. of students graduated from the program (n) = 187

Success Rate: n/m * 100 = 88.20

No.of students admitted in corresponding first year (PG) (m1) = Nil

No. of students graduated from the program (n1) = Nil

PG Success Rate: n1 /m1*100 = Nil

Students Placements, Higher Studies, Entrepreneurships No. of students who are placed =130

No. of students went for higher studies = 6

No. of students with multiple offers = 71

Highest and mean packages received by the students = 10 LPA & 4.16 LPA

No. of students with entrepreneurship details = 0

No. of students who qualified in competitive examinations (Civil Services, GATE, GMAT, CAT, GRE, TOEFL, etc.) = 0

Students Activities

No. of Award -Winning Students in Technical events:

S.No	Name of the Student	Name of the Technical Event	Institutional / State / National/ International)	Organized By	Date(s)	Achievements & Recognitions
1	G. Bindu Prasanna	Individual	College	Annual Day	05.04.2025	Best Student
2	k. Madhuri	Individual	College	Graduation Day	23.08.2025	Topper in CSE

No. of Students Participated in Technical Events:

S.No.	Name of the Student	Name of the Technical Event	Institutional / State / National / International)	Organized By	Date(s)
1	Uppathala Laxman	SRM Smart Park Hackathon	State	SRM Institute of Science & Technology	28-10-2025
2	Cherukuri Madhu	SRM Smart Park Hackathon	State	SRM Institute of Science & Technology	28-10-2025
3	A. Siri chandana	Viksit Bharat Youth Parliament	State	Anurag University, Hyderabad	28.03.2025 & 29.03.2025
4	N.Anusha	Aksharamala	State	IIT Hyderabad	29.08.2025
5	Sk.Afra	Aksharamala	State	IIT Hyderabad	29.08.2025

6	P.Gayatri	Aksharamala	State	IIT Hyderabad	29.08.2025
7	N.Ramya Sri	Aksharamala	State	IIT Hyderabad	29.08.2025
8	P.Durga Bhavani	Aksharamala	State	IIT Hyderabad	29.08.2025
9	D.Ramya	Aksharamala	State	IIT Hyderabad	29.08.2025
10	S.Navyasri	Aksharamala	State	IIT Hyderabad	29.08.2025
11	V.Pavan Sai	Aksharamala	State	IIT Hyderabad	29.08.2025
12	K.Sandeep	Aksharamala	State	IIT Hyderabad	29.08.2025
13	k. sai sharan	Aksharamala	State	IIT Hyderabad	29.08.2025
14	N. Gnanendra	Aksharamala	National	India Post Payments Bank	12.11.2024
15	R. Praveen	Aksharamala	National	India Post Payments Bank	12.11.2024
16	Shaik Nazeem	Aksharamala	National	India Post Payments Bank	12.11.2024
17	C.S.V.Vaishnavi Devi	Webinar on Data Analytics	Online	Novitech	21.07.2025
18	C.S.V.Vaishnavi Devi	Seminar	Online	Dexterity Edtech Private Limited	14.08.2025
19	C.S.V.Vaishnavi Devi	Seminar on Quantum Computing	Online	Quality Thought	23.11.2025
20	C.S.V.Vaishnavi Devi	AI Bootcamp Webinar	Online	Team Freedom with AI	25.11.2025

21	C.S.V.Vaishnavi Devi	Build Career Strong – Webinar	Online	Grand Guru	10.12.2025
22	C.S.V.Vaishnavi Devi	From Idea to Publication: A Beginner's Guide for UG Researchers	Online	Gudsky Research Foundation	20.12.2025
23	Sumayyah Anjum	Work Shop	Online	Brainic Career Guidance	18.10.2025
24	Sumayyah Anjum	Codequezt #25 Coding Challenge	Online	Naukri Campus	30.11.2025
25	Sumayyah Anjum	Brandquezt # 25 , Crossovers and Co - Branding	Online	Naukri Campus	16.11.2025
26	Sumayyah Anjum	Tata Crucible Campus Quiz	Online	Tata Group	28.11.2025
27	Sumayyah Anjum	Brandquezt #26 The Great Rebrands	Online	Naukri Campus	12.12.2025
28	Sumayyah Anjum	Techquezt #26 AI all Round Us	Online	Naukri Campus	21.12.2025
	Sumayyah Anjum	Crack Government Exam - Bank PO #3	Online	Naukri Campus	21.12.2025

No. of Award -Winning Students in Sports/Cultural Events:

S. No	Name of the Student	Event Category (Sports / Cultural)	Institutional / State / National / International)	Organized By	Date(s)	Achievements & Recognitions

1	B Mohan Phanindra	Throw Ball	National	Telangana Throw Ball Association	11.11.2024	Bronze Medal (Runner)

No. of Students Participated in Sport/Cultural Events:

S.No.	Name of the Student	Event Category (Sports/Cultural)	Institutional / State / National / International)	Organized By	Date(s)
1	N. Sri Ajay	Sports (Kabaddi)	National	Anurag University, Hyderabad	19-12-2025 To 20-12-2025
2	N. Sri Ajay	Sports (Kabaddi)	National	Sai Kabaddi Association, Karimnagar	25-12-2025 To 28-12-2025
3	N. Sri Ajay	Sports (Kabaddi)	National	South Zone University Kabaddi Championship, Karnataka	03-10-2025 To 06-10-2025
4	C.S.V.Vaishnavi Devi	Motivational Talk for B.Tech Students	Institutional	Kits College	04.o4.2024

No. of Students Participated in NSS Activities:

S.No	Name of the Student (Beneficiaries)	NSS Activity / Programme Name	Institutional / State / National/ International)	Organized By	Date(s)

1	106	International Yoga day Celebrations	State	NSS UNIT	21.06.2024
2	440	An Awareness Program on Menstrual Health & Hygiene in Girls	State	NSS UNIT in Association With “Stambhadhri Hospital”, Khammam	03.08.2024
3	473	An Awareness Program on Gynaecological Cancers in Women	State	NSS UNIT in Association With “American Oncology Institute”, Hyderabad	25.09.2024
4	04	Heartfulness Essay Event 2024	National (Online)	Heartfulness Education Trust and Shri Ram Chandra Mission in Collaboration with The Commonwealth	Before 30.08.2024
5	92	Swachhata Hi Seva Observance Campaign In Bojjagudem Thanda Village	State	NSS UNIT under “Ministry of Education”, Govt. of India	30.09.2024
6	387	Swachhata Hi Seva Observance Campaign in the Campus	State	NSS UNIT under “Ministry of Education”, Govt. of India	30.09.2024
7	215	Bathukamma Celebrations	State	NSS UNIT	05-10-2024
8	36	75th Constitutional Day Celebrations-Essay Writing Competition	State	NSS UNIT in Association with “Central Bureau of Communication”-Field Office-Nalgonda	25.11.2024
9	279	75th Constitutional Day Celebrations	State	NSS UNIT in Association with “Central Bureau of	26.11.2024

				Communication"- Field Office- Nalgonda	
10	117	Sankranti Sambaralu	State	NSS UNIT	10.01.2025
11	160	An Awareness Program on Road Safety	State	NSS UNIT in Association with " RTA Unit Office" Kodad	28.012025
12	179	An Awareness Program on Eradication of Drugs	State	NSS UNIT in Association with "Police Department, Kodad Division"	28.01.2025
13	142	Blood Donation for Thalassemia Children	State	NSS UNIT in Association with "Thalassemia Sickle Cell Society"- Nova Care Blood Bank Khammam	07.02.2025
14	48	An Awareness Program on Menstrual Health & Hygiene and Gynaecological Cancers in Girls	State	NSS UNIT	24.02.2025
15	172	Free Medical Camp in Ananthagiri Village	State	NSS UNIT under Assistance from "Aruna Hospital Kodad and Big TV"	02.03.2025
16	121	Tercentenary Birth Celebrations of Lokamata Devi Ahilyabai Holkar	State	NSS UNIT in Association with **"**Akhil Bhartiya Rashtyira Shaikshik Mahasangh"	03.03.2025

17	08	My Bharat - Youth Parliament- District & State Level	State	"Ministry of Youth Affairs and Sports", Govt. of India	21st, 28th & 29th March 2025
18	22	Aksharamaala-Online Educational Services	State (ONLINE)	IIT Hydeabad	Academic Year 2024-2025

6. **Curriculum Development**

New Courses

UG:

No.of new courses introduced (Mention the Regulation) = 5

No. of courses with change in syllabus (if more than 20%) = Nil

PG:

No. of new courses introduced (Mention the Regulation) = Nil

No.of courses with change in syllabus (ifmorethan20%) = Nil

Details of Conduction of Remedial Classes for Academically Weak Students:

S.No.	Name of the Course	Name of the Faulty	Duration of the Course From (Date)To(Date)	No. of Classes taken	No. of Weak Students	No. of Students attended	No. of Students passed
1	Programming for Problem Solving	Mr K.Naresh kumar	17.03.2025 To 17.04.2025	20	16	16	14
2	Programming for Problem Solving	Mr. V.Jeevan Babu	17.03.2025 To 17.04.2025	18	22	22	16

3	Programming for Problem Solving	Mr. V.Jeevan Babu	17.03.2025 To 17.04.2025	22	28	28	15

No. of Value-Added Courses Conducted:

S.No.	Name of the Value Added Course	Duration of the Course	No. of Hours	Faculty Coordinator	Collaborative Industry	No. of Students Registered	No. of Students Passed
1	Javascript & React.js	12 Weeks	33	P. Kishore	Anurag Engineering College	80	80
2	SQL,PL/SQL	12 Weeks	30	Ch.Krishna Prasad	Anurag Engineering College	75	75
3	HTML& CSS	12 Weeks	30	J. Vijayasree	Anurag Engineering College	78	78

No. Students Activities Organized by the Department:

S.No	Name of the Activity	Type of Activity (Technical / Cultural / Sports / Extension)	Date(s)	Level(Department/Institution/ State / National)	No. of Students Participated	Organizing Committee / Coordinator	Online / Offline
1	Bathukamma Celebrations	Cultural	05-10-2024	Institution	1500+	Mrs. M. Anusha	Offline
2	Vaibhavam AI	Technical	24-09-2024	Department	160	Mrs. J.Vijayasree	Offline

3	Two Days Workshiop On Engineering With Aws	Technical	26-09-2024	Department	155	Dr. Ch.N. Santhosh Kumar	Offline
4	Campus Recruitment Training Program	Technical	26-09-2024	Department	162	Mr. P.Kishore	Offline
5	Hands On Workshop On Block Chain Technology	Technical	05-11-2024	Department	150	Dr. Ch.N. Santhosh Kumar	Offline
6	Report On Hands On Workshop On Data Analytics With AWS	Technical	04-11-2024	Department	164	Dr. Ch.N. Santhosh Kumar	Offline
7	Sankranthi Celebration	Cultural	10-01-2025	Institution	1657	Mrs.G. Prashanthi	Offline
8	Full Stack Development	Technical	09-02-2025	Department	80	Mrs.J. Vijayasree	Offline
9	International Womens Day Celebrations	Cultural	08-03-2025	Institution	55	Mrs. M Anusha	Offline
10	Full Stack Development	Technical	09-04-2025	Department	80	Mrs. J Vijayasree	Offline

Suggestions for the areas of improvement:

Name of the Audit member	Designation	Signature	Date
Dr. G. Charles Babu	Professor, GRIET, Hyderabad		12/9/25
Dr M. G Sethuraman	Sr. Professor		12/9/25

IQAC Coordinator
Anurag Engineering College
Ananthagiri (V&M) Kodad,
Suryapet (Dt.), Telangana-508206.

PRINCIPAL
PRINCIPAL
Anurag Engineering College
Ananthagiri (V&M), KODAD.
Suryapet (Dt.), Telangana-508206

8.3 Improvement in Faculty Qualification/Contribution (15) — **Total Marks 15.00**

Institute Marks : 15.00

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
No. of faculty members with Ph.D. degree	9.00	7.00	7.00
No. of publications in peer reviewed journals	4.00	4.00	2.00
No. of publications in conferences	9.00	2.00	0.00

8.4 Improvement in Academic Performance (10) — **Total Marks 10.00**

Institute Marks : 10.00

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)

Academic Performance Index (API) of the First-Year Students in the Program (Refer to section 4.3)	7.23	7.00	6.69
Academic Performance Index (API) of the Second-Year Students in the Program (Refer to section 4.4)	6.77	7.00	6.94
Academic Performance Index (API) of the Third-Year Students in the Program (Refer to section 4.5)	7.78	7.71	7.72